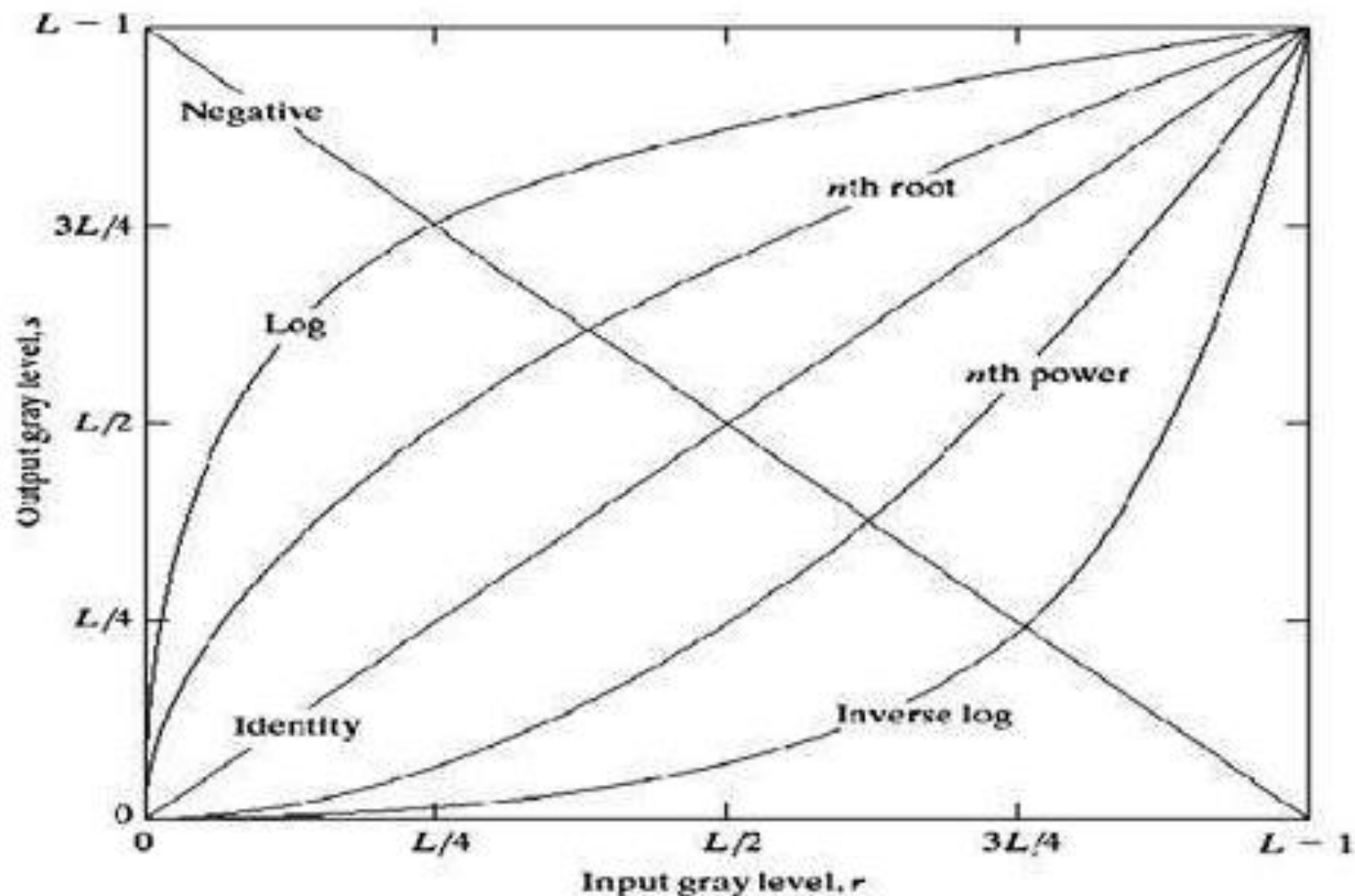


IMAGE PROCESSING AND ITS APPLICATION

MODULE - 2

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Assistant Professor
Dept. of Robotics and Artificial
Intelligence

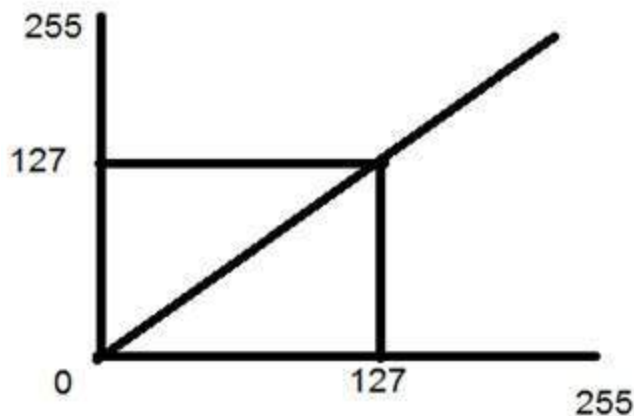
Basic Grey level transforms



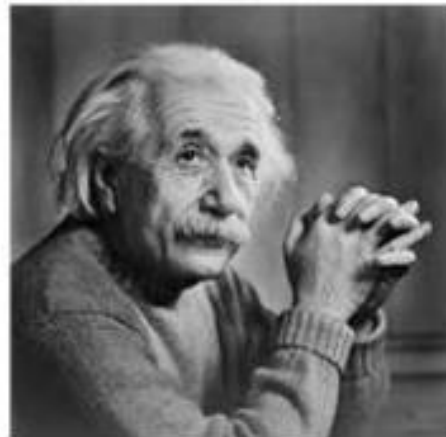
Basic Grey level transforms

Identity transition

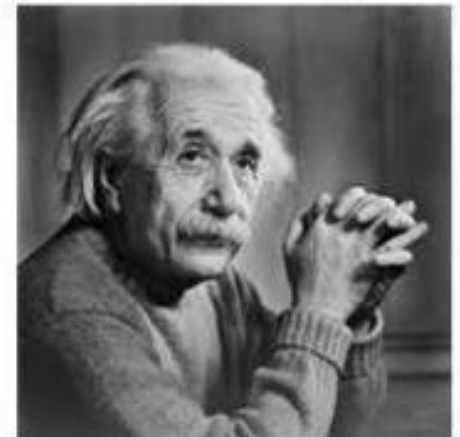
- Identity transition is shown by a straight line.
- Each value of the input image is directly mapped to each other value of output image.
- That results in the same input image and output image.



Input image



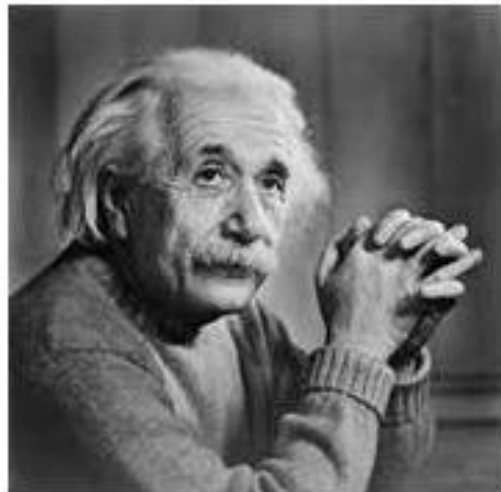
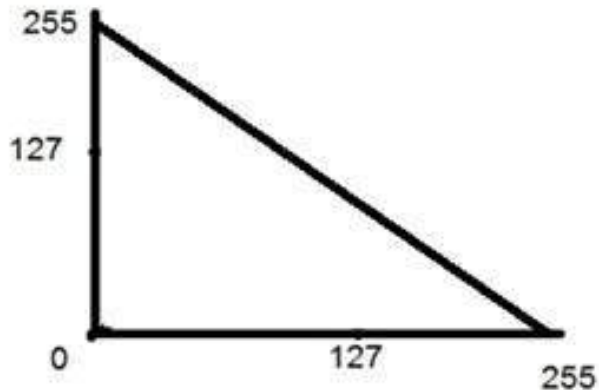
Output image



Basic Grey level transforms

Negative transformation

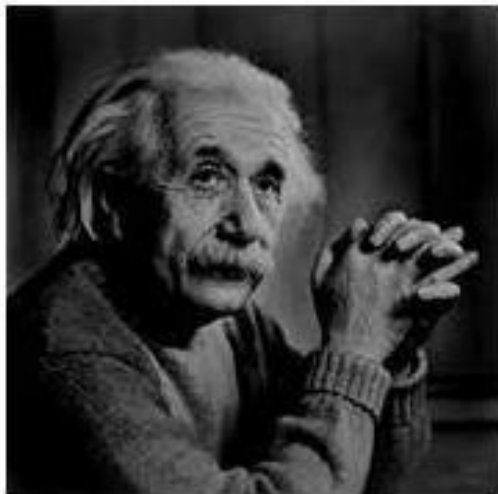
- Negative transformation, is invert of identity transformation.
- In negative transformation, each value of the input image is subtracted from the L-1 and mapped onto the output image.



Basic Grey level transforms

Log transformation

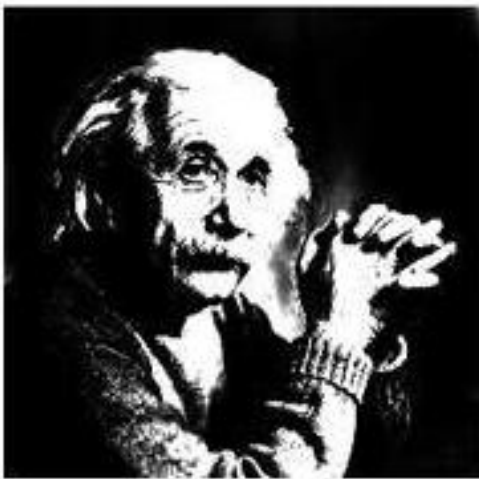
- $s = c \log(r + 1)$.
- Where s and r are the pixel values of the output and the input image and c is a constant.
- The value 1 is added to each of the pixel value of the input image because if there is a pixel intensity of 0 in the image, then $\log(0)$ is equal to infinity.
- So 1 is added, to make the minimum value at least 1.



Basic Grey level transforms

Power Law transformations

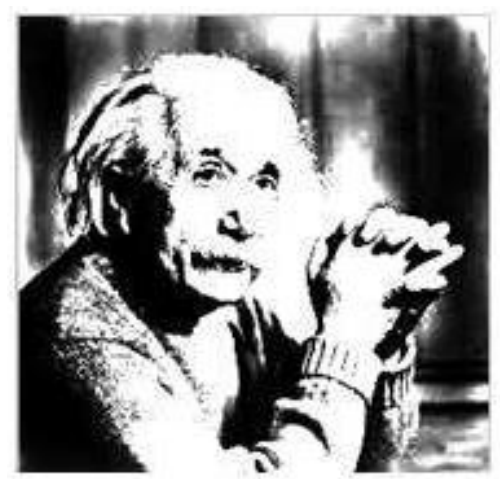
- There are further two transformation is power law transformations, that include nth power and nth root transformation.
- $s = cr^\gamma$
- This symbol is called gamma, due to which this transformation is also known as gamma transformation.



Gamma = 10



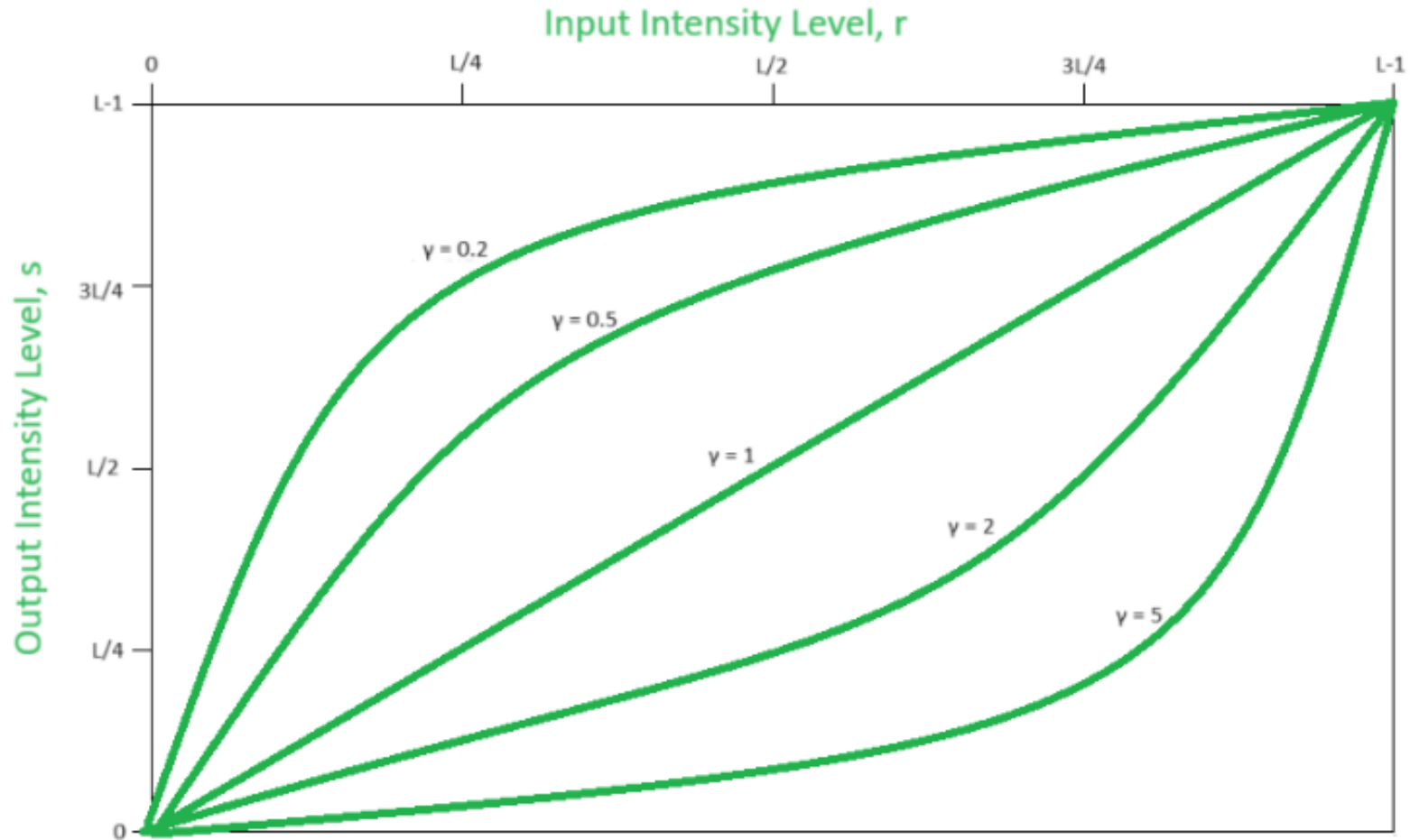
Gamma = 8



Gamma = 6

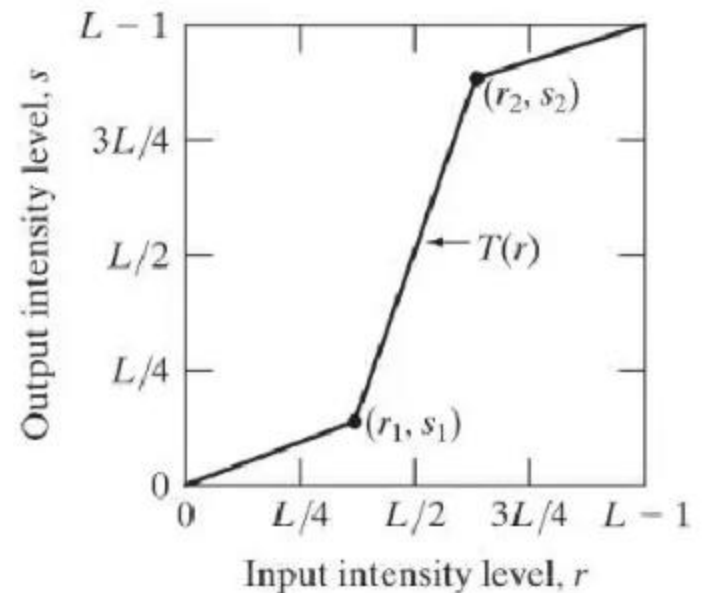
Basic Grey level transforms

Power Law transformations



Piece-wise Linear Transformation

- **Piece-wise Linear Transformation** is type of gray level transformation that is used for image enhancement.
- It is a spatial domain method. It is used for manipulation of an image so that the result is more suitable than the original for a specific application.
- Some commonly used piece-wise linear transformations are: **Contrast Stretching**



Histogram processing

- Histogram processing in digital image processing is a foundational technique that analyzes and modifies the statistical distribution of pixel intensities to enhance image contrast and visual quality.

Histogram processing

- Primary Techniques
- **Histogram Equalization (Global Enhancement):** Aims to transform the intensity levels to create a uniform, flat histogram, enhancing contrast across the entire image. It uses the Cumulative Distribution Function (CDF) to map old gray levels to new ones.
- **Histogram Matching (Specification):** Modifies the image's histogram to match a specific, desired shape, which is useful for enhancing specific, local features or for comparing images.
- **Local Enhancement:** Applies transformation functions based on the intensity distributions of local neighborhoods (small areas) rather than the global image, which is better for bringing out details in specific, small, or low-contrast regions

Histogram processing

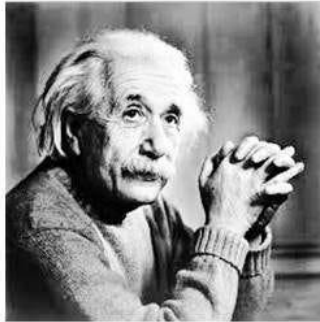
- Common Applications
 - **Contrast Enhancement:** Improving the visibility of details in low-contrast images.
 - **Image Thresholding:** Separating foreground from background, especially for image segmentation.
 - **Brightness Adjustment:** Correcting images that are too dark or too bright.
 - **Image Analysis:** Analyzing statistical properties to determine exposure or scene brightness.

Histogram equalization

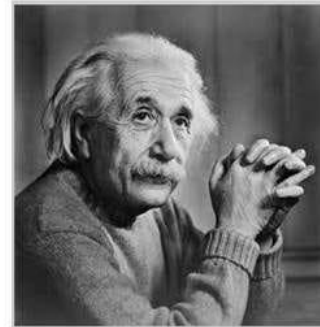
- **Histogram equalization** is a **contrast enhancement technique** that redistributes the gray-level intensities of an image so that the output image has a **more uniform histogram**. This improves visibility of details, especially in images that are too dark or too bright
- Histogram equalization is a widely used, computationally simple image processing technique that enhances contrast by spreading out the most frequent intensity values, effectively stretching the intensity range of an image to be more uniform.

Histogram equalization

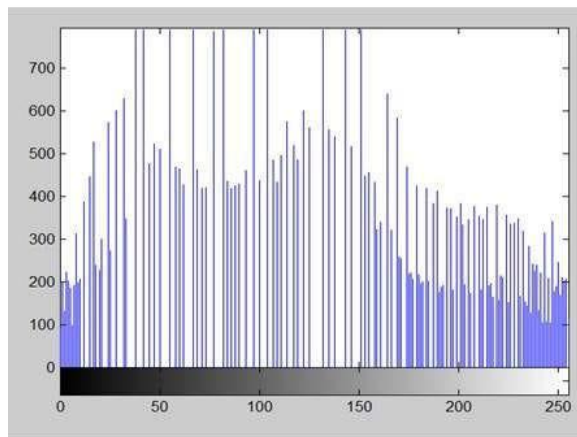
New Image



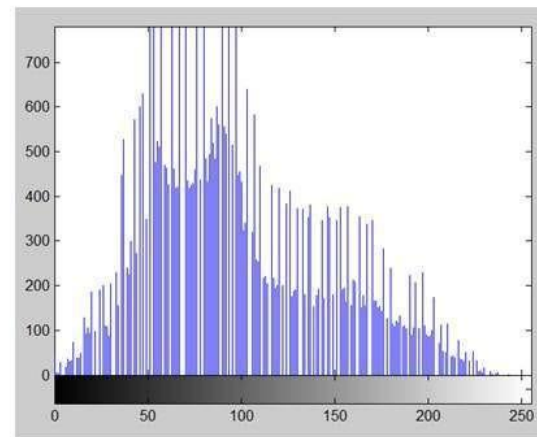
Old image



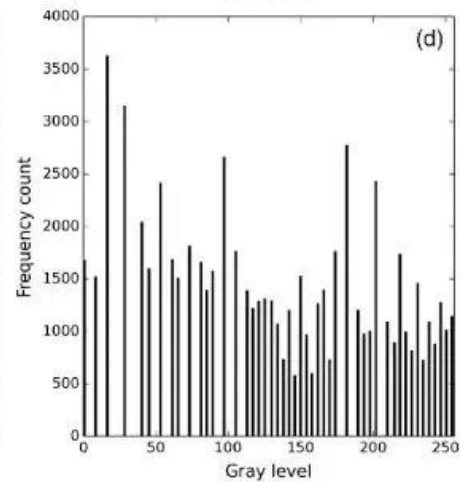
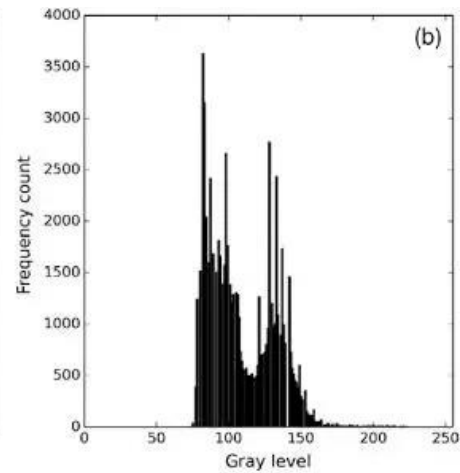
New Histogram



Old Histogram



Histogram equalization



Why Histogram Equalization?

- Many images use only a **small portion of the available intensity range** (0–255). As a result:
 - Details are hidden in dark or bright regions
 - Contrast is poor
- Histogram equalization **stretches and redistributes** intensity values across the full range.

Histogram equalization (working)

- Compute the histogram of the image
- Convert it to a **cumulative distribution function (CDF)**
- Use the CDF as a **mapping function** to remap old gray levels to new ones
- The transformation is **global** (same mapping for all pixels).

Histogram equalization

Advantages

- Simple and effective contrast enhancement
- Fully automatic (no parameters required)
- Improves global contrast

Limitations

- May over-enhance noise
- Not suitable for images with already good contrast
- Global method → ignores local details
- Can produce unnatural appearance

Histogram equalization

- **Applications**

- Medical imaging (X-ray, MRI)
- Satellite and aerial images
- Document image enhancement
- Industrial inspection

Smoothing Frequency Domain

- **Smoothing frequency domain filters** are used to **reduce noise and fine details** in an image by **attenuating high-frequency components** while allowing low-frequency components to pass.
- In the frequency domain:
 - **Low frequencies** → smooth regions
 - **High frequencies** → edges, noise, sharp transitions

Smoothing Frequency Domain

Frequency Domain Filtering Model

1. Compute Fourier transform of image

$$F(u, v) = \mathcal{F}\{f(x, y)\}$$

2. Multiply with filter transfer function

$$G(u, v) = H(u, v) F(u, v)$$

3. Compute inverse Fourier transform

$$g(x, y) = \mathcal{F}^{-1}\{G(u, v)\}$$

- $f(x, y) \rightarrow$ input image
- $H(u, v) \rightarrow$ smoothing filter
- $g(x, y) \rightarrow$ smoothed image

1. Ideal Low-Pass Filter (ILPF)

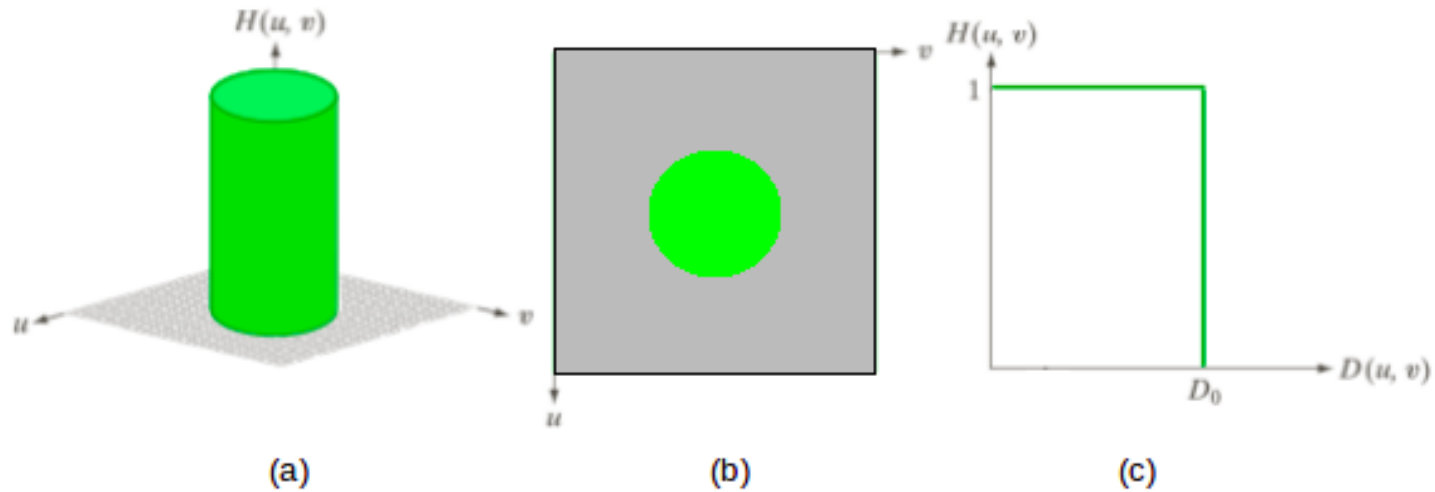
Passes all frequencies inside a cutoff radius and blocks others.

$$H(u, v) = \begin{cases} 1 & D(u, v) \leq D_0 \\ 0 & D(u, v) > D_0 \end{cases}$$

Where:

- $D(u, v)$ = distance from origin of frequency plane
- D_0 = cutoff frequency
- Sharp cutoff
- Produces **ringing effect**

1. Ideal Low-Pass Filter (ILPF)



(a): Perspective plot of an ideal lowpass filter transfer function, (b): Filter displayed as an image, (c): Filter radial cross-section



2. Butterworth Low-Pass Filter (BLPF)

Provides a smoother transition between passband and stopband.

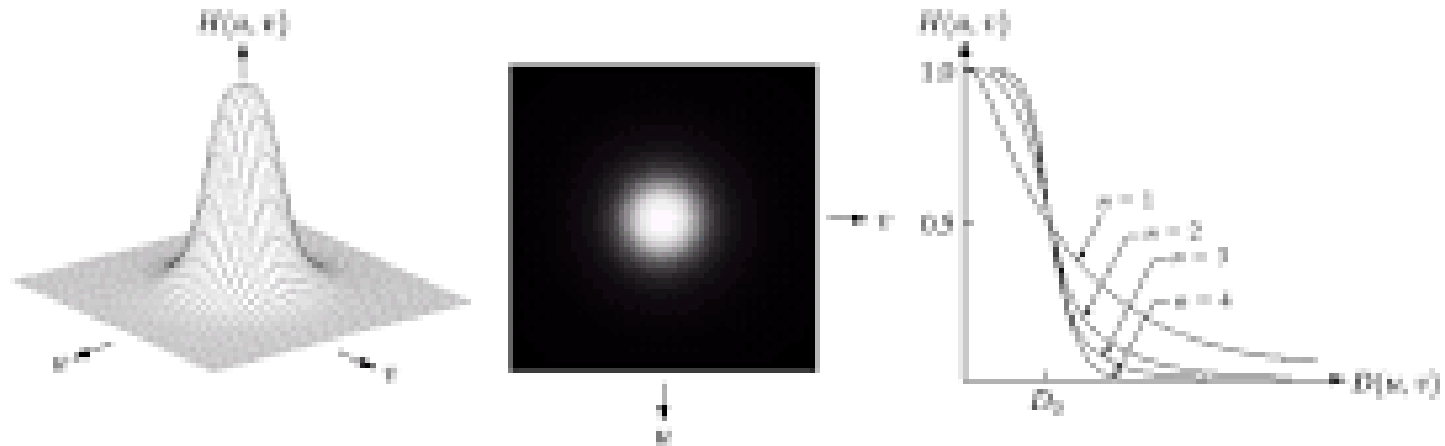
Transfer Function

$$H(u, v) = \frac{1}{1 + \left(\frac{D(u, v)}{D_0} \right)^{2n}}$$

Where:

- n = order of the filter
- No sharp discontinuity
- Higher order $\rightarrow$ sharper cutoff
- Less ringing than ILPF

2. Butterworth Low-Pass Filter (BLPF)



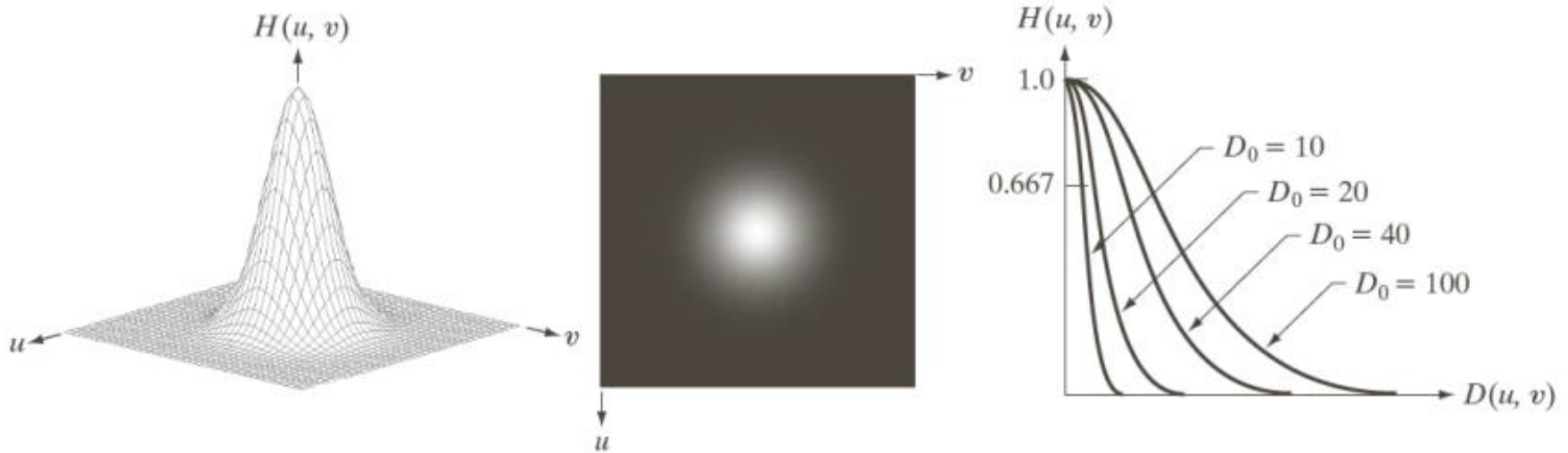
3. Gaussian Low-Pass Filter (GLPF)

- Uses a Gaussian function for smooth filtering.

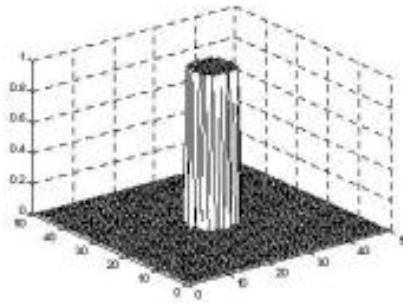
$$H(u, v) = e^{-\frac{D^2(u, v)}{2D_0^2}}$$

- Smoothest response
- No ringing
- Best overall smoothing behavior

3. Gaussian Low-Pass Filter (GLPF)



Smoothing Frequency Domain

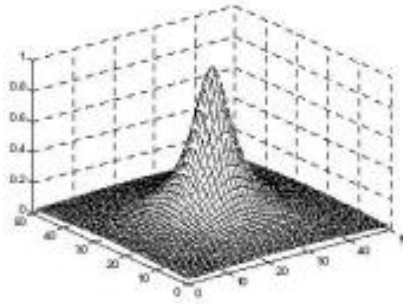


(a)



(b)

Ideal LPF

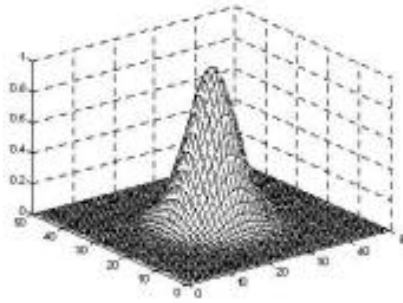


(c)



(d)

Gaussian LPF



(e)



(f)

Butterworth LPF

Smoothing Frequency Domain

Effects of Smoothing in Frequency Domain

- Reduces noise
- Blurs edges
- Removes fine details
- Improves overall smoothness

Applications

- Noise reduction
- Preprocessing for segmentation
- Medical image smoothing
- Satellite image enhancement

Smoothing Frequency Domain

Advantages

- Efficient for large kernels
- Easy control over frequency components
- Precise filtering

Disadvantages

- Requires Fourier transform
- Not suitable for real-time low-power systems
- Global filtering (no spatial adaptivity)

Sharpening Frequency Domain Filters

- Sharpening frequency domain filters are used to enhance edges, fine details, and abrupt intensity changes in an image by emphasizing high-frequency components.
- Sharpening filters suppress low frequencies and amplify high frequencies.

Sharpening Frequency Domain Filters

1. Compute Fourier transform:

$$F(u, v) = \mathcal{F}\{f(x, y)\}$$

2. Apply sharpening filter:

$$G(u, v) = H(u, v) F(u, v)$$

3. Inverse Fourier transform:

$$g(x, y) = \mathcal{F}^{-1}\{G(u, v)\}$$

- $f(x, y) \rightarrow$ input image
- $H(u, v) \rightarrow$ sharpening filter
- $g(x, y) \rightarrow$ sharpened image

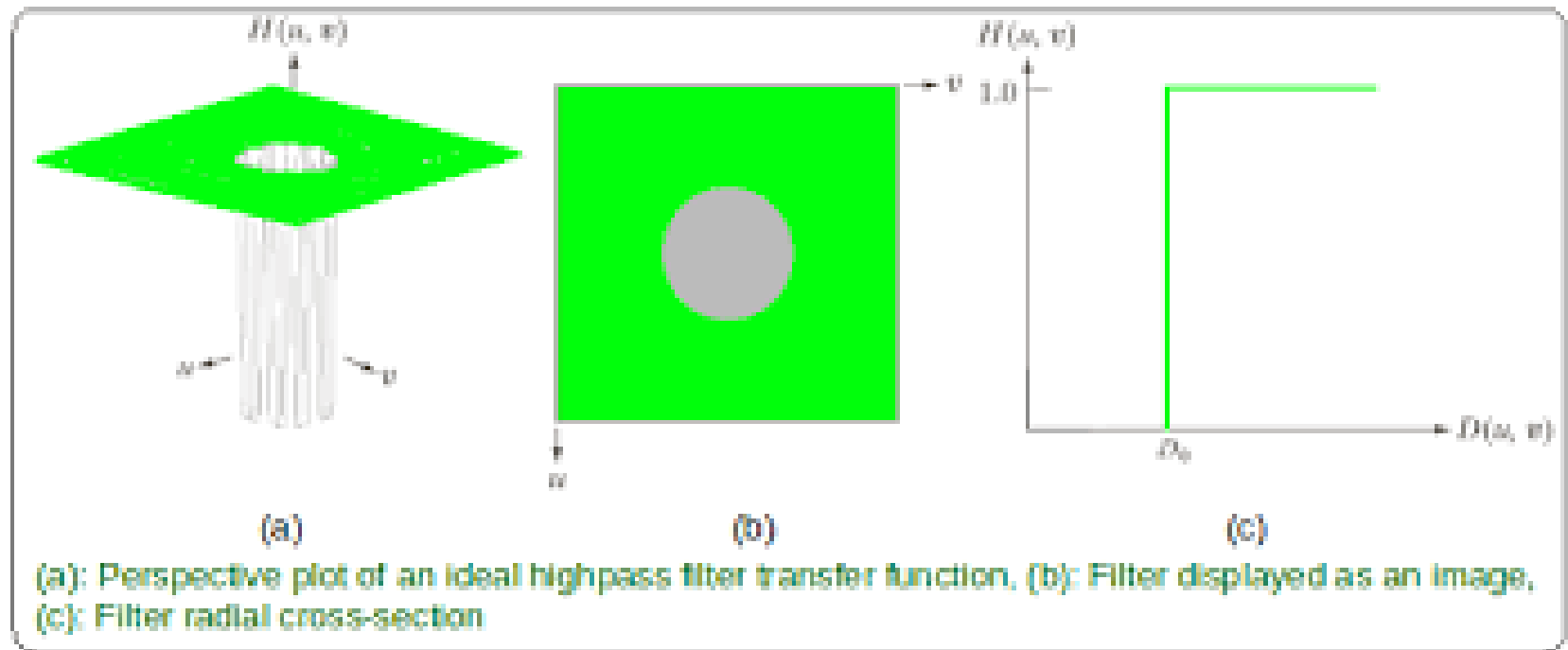
1. Ideal High-Pass Filter (IHPF)

$$H(u, v) = \begin{cases} 0 & D(u, v) \leq D_0 \\ 1 & D(u, v) > D_0 \end{cases}$$

Characteristics

- Sharp cutoff
- Strong edge enhancement
- Severe **ringing effect**

1. Ideal High-Pass Filter (IHPF)



2. Butterworth High-Pass Filter (BHPF)

Provides a smoother transition between low and high frequencies.

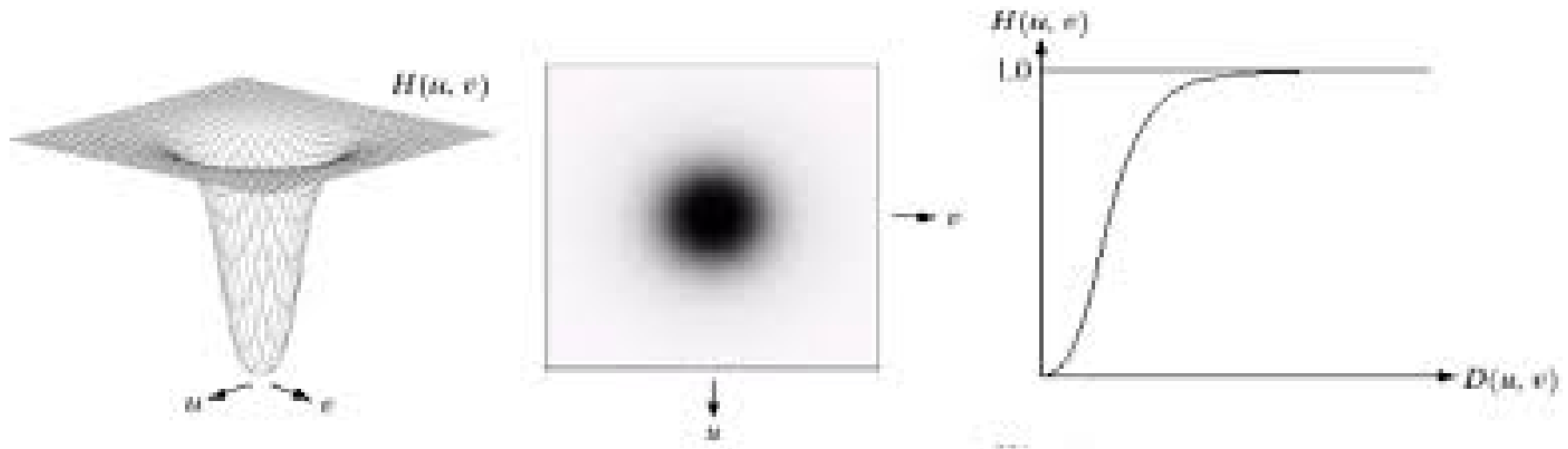
Characteristics

- Controlled sharpening
- Less ringing than IHPF
- Higher order $\rightarrow$ stronger sharpening

$$H(u, v) = \frac{1}{1 + \left(\frac{D_0}{D(u, v)} \right)^{2n}}$$

$n \rightarrow$ filter order

2. Butterworth High-Pass Filter (BHPF)

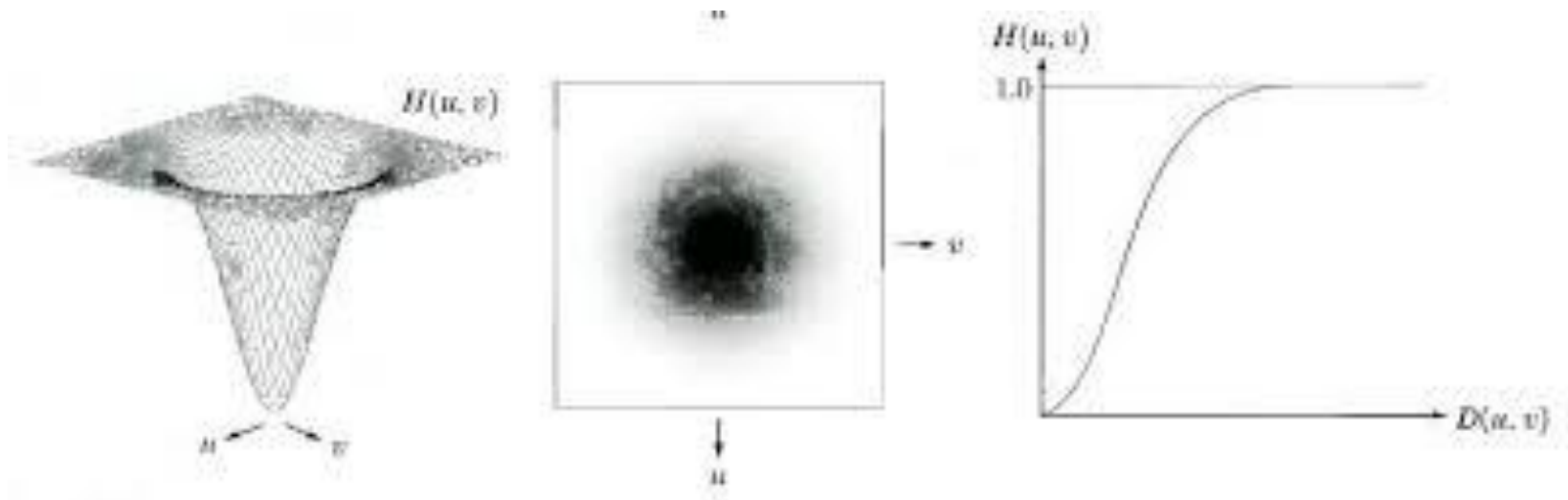


3. Gaussian High-Pass Filter (GHPF)

- Uses Gaussian function to smoothly suppress low frequencies.
- **Characteristics**
 - Smoothest sharpening
 - No ringing artifacts
 - Best practical choice

$$H(u, v) = 1 - e^{-\frac{D^2(u, v)}{2D_0^2}}$$

3. Gaussian High-Pass Filter (GHPF)



Sharpening Frequency Domain Filters

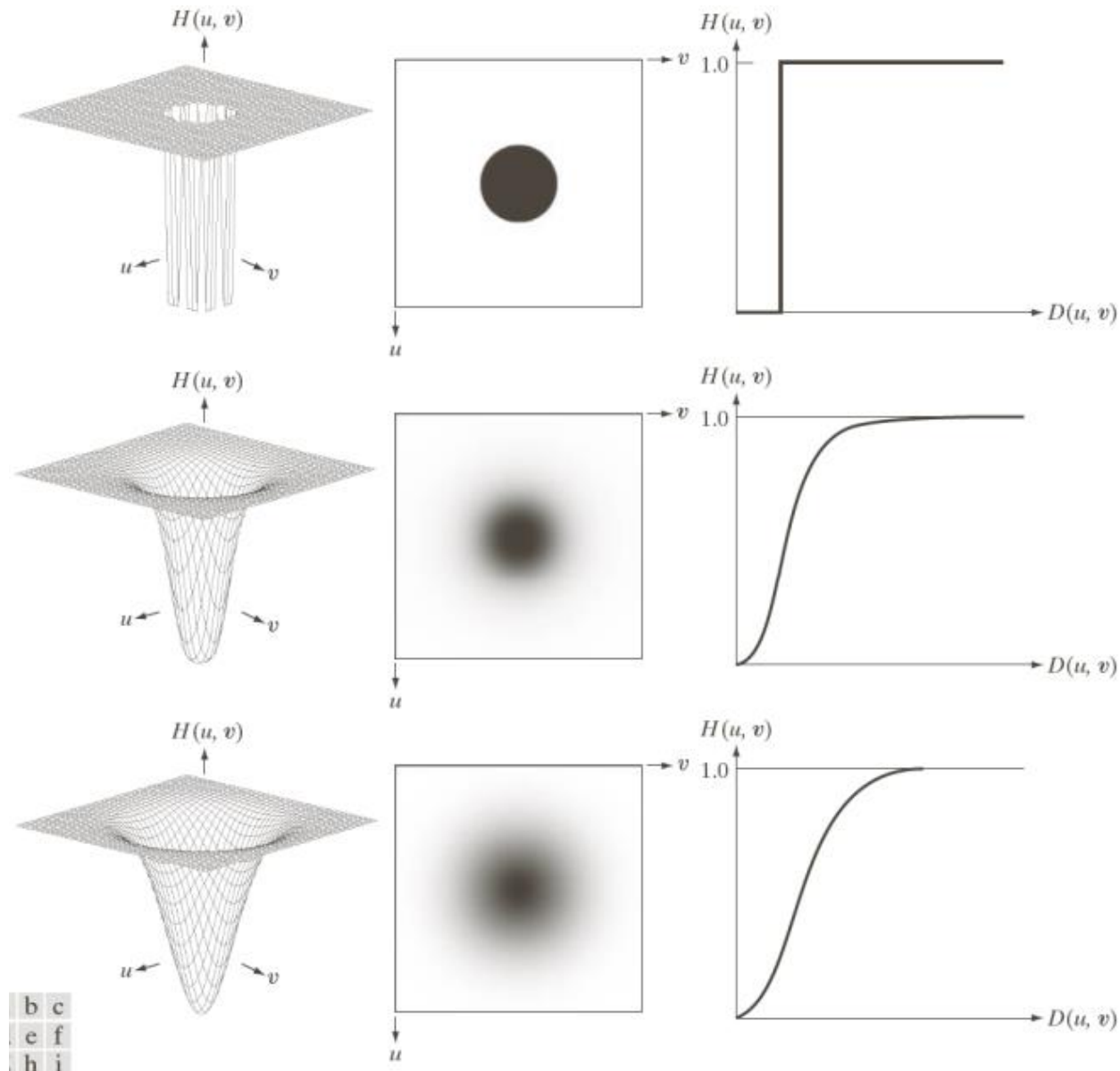


Image Degradation and Restoration Process

- During image acquisition, transmission, or storage, images often suffer from **degradation** due to noise, blur, or system imperfections.
- **Image restoration** aims to **recover the original image** by modeling and reversing the degradation process.

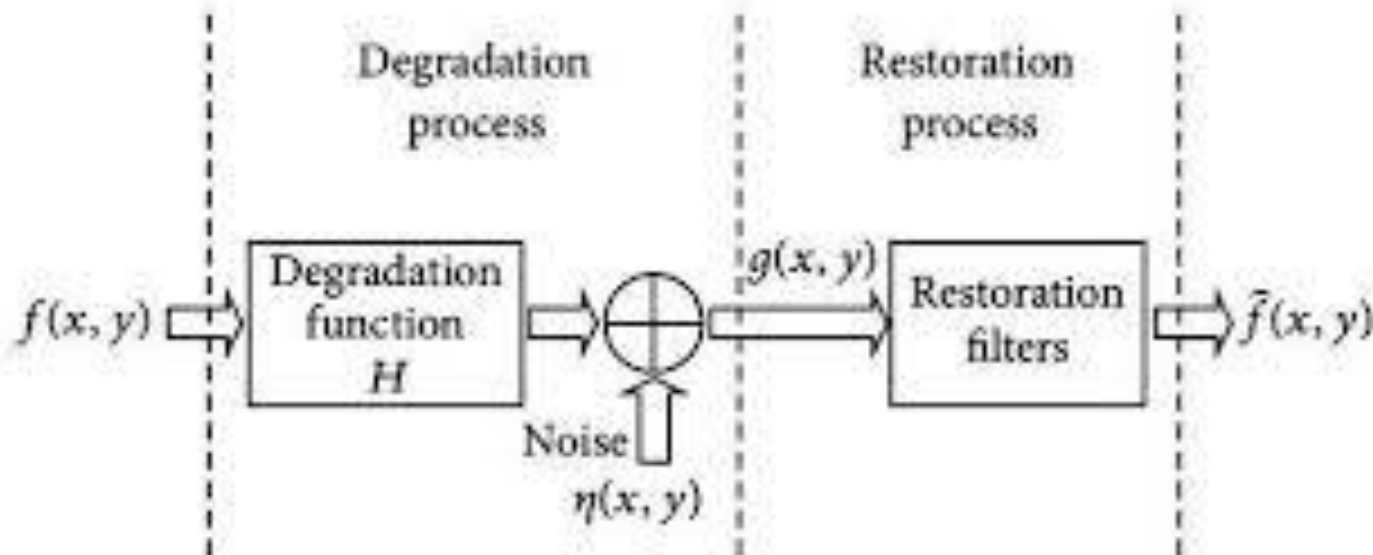


Image Degradation Process

Image degradation is the distortion of an image caused by external factors such as noise, motion, or defocusing.

Causes of Image Degradation

- **Noise**
 - Gaussian noise
 - Salt and pepper noise
 - Poisson noise
- **Blur**
 - Motion blur
 - Defocus blur
 - Atmospheric turbulence
- **Sampling and Quantization errors**

Image Restoration Process

- **Image restoration** is the process of estimating the original image from a degraded image using **mathematical and probabilistic models**.
- Restoration Techniques
 - Inverse Filtering
 - Wiener Filtering
 - Least Squares Filtering

Image Degradation and Restoration Process

Applications

- Medical imaging
- Astronomical image correction
- Satellite image restoration
- Old photograph restoration

Image Degradation and Restoration Process

Advantages

- Recovers true image details
- Model-based approach
- Suitable for scientific applications

Limitations

- Requires accurate degradation model
- Computationally expensive
- Sensitive to incorrect assumptions

Noise Models in Image Processing

- **Noise** is any unwanted disturbance that degrades the quality of an image.
- A **noise model** mathematically describes the statistical behavior of noise introduced during image acquisition, transmission, or processing.

$$g(x, y) = f(x, y) + \eta(x, y)$$

$f(x, y) \rightarrow$ original image

$\eta(x, y) \rightarrow$ noise

$g(x, y) \rightarrow$ noisy image

Noise Models in Image Processing

Original image



Gaussian noise



Poisson noise



Salt and pepper noise



Gaussian Noise

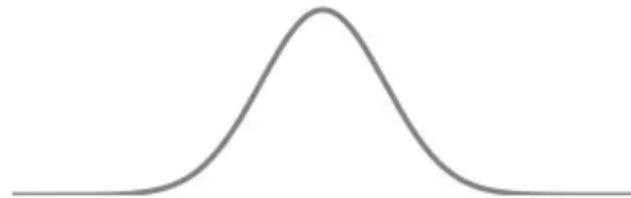
- Image noise is referred to as Gaussian if the random fluctuations follow a normal (Gaussian) distribution
- Caused due to various properties of the sensor including its temperature and level of illumination

Characteristics

- Affects all pixels
- Common in electronic circuits

Removal Technique

- Mean filter
- Wiener filter



Gaussian Noise

Original Image



Gaussian Noise

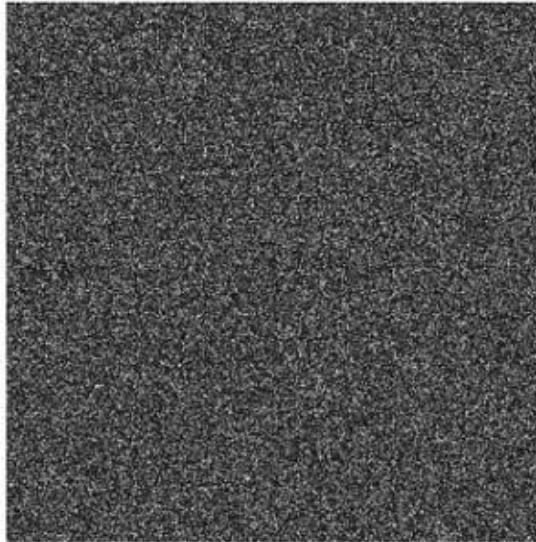


Image with Gaussian Noise



Salt and Pepper Noise

- If the random fluctuations appear suddenly and unexpectedly (hence the name impulse) in random portions of an image.
- Impulse noise is also known as salt and pepper noise since these noise values are either black(pepper) or white(salt) pixel values.
- This type of noise is caused due to sharp and sudden disturbances in the image signal, and is mainly generated by errors in analog to digital conversion or bit transmission.
- Other types of noise are introduced by adding or multiplying noise values to each pixel in an image. Impulse noise, however, is introduced by replacing a pixel value independently by either a minimum (0) or a maximum (255).

Salt and Pepper Noise

Original Image



Salt and Pepper Noise



Image with Salt and Pepper Noise



Uniform Noise

- Noise in an image is referred to as uniform noise if the random fluctuations are distributed evenly within an image.
- It is also known as quantization noise since it comes about when the values of a captured image are mapped to a set of discrete values.
- It appears often in situations of poor lighting

Uniform Noise

Original Image



Uniform Noise

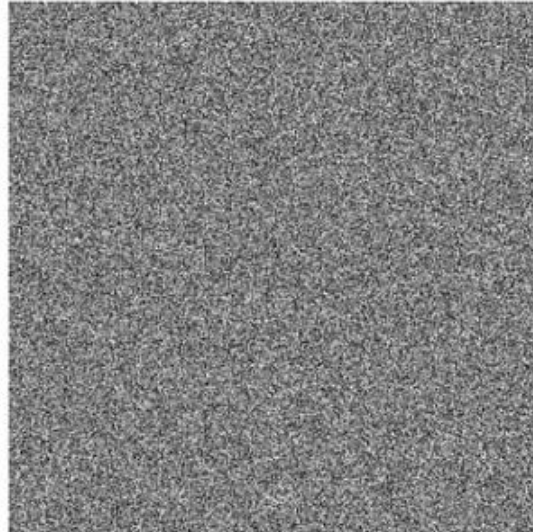


Image with Uniform Noise



Image Filtering

- Filtering techniques are use to enhance and modify digital images.
- Images filters are use to blurring and noise reduction , sharpening and edge detection.
- Image filters are mainly use for suppress high (smoothing techniques) and low frequencies (image enhancement, edge detection).

Median Filter

- Median filter is a non-linear filter.
- It replaces each pixel values by the median values of it's neighbor pixels.
- This is the efficient way for remove salt-and-pepper noise.

30	35	40	42	42
35	42	37	37	40
38	39	40	41	42
40	41	42	43	43
42	43	45	44	46

		41		

37, 37, 39, 40, **41**, 42, 42, 43

Median Filter



Mean Filtering

- In average filtering, pixel values will be replaced by average values of neighbour pixels.
- A kernel moves across the image computing the average of each pixel
- Used to reduce gaussian and uniform noise
- It blurs the image and reduces sharpness

23	25	30	35	30
25	30	35	37	40
45	40	37	43	45
38	40	43	42	46
35	40	42	45	47

		39		

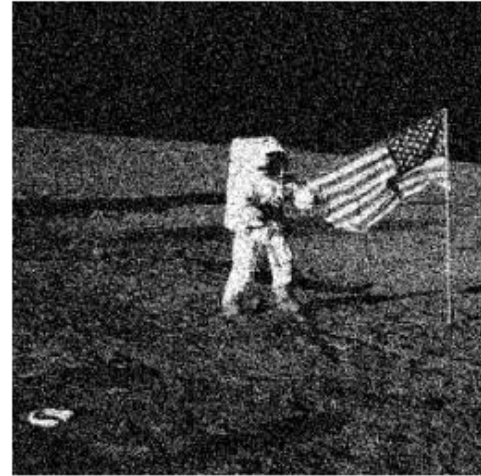
$$\text{Value} = (30 + 35 + 37 + 40 + 37 + 43 + 40 + 43 + 42) / 9 = 38.55 = 39$$

Mean Filtering

Original



With Noise



Mean_Filter



Mean_Filter Geometric



Laplacian Filter

- Laplace smoothing technique is mainly use to detect image edges.
- It highlights gray level discontinuities.
- It is based on second spatial derivation of an image.

$$Laplace(f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Laplacian Filter

- Laplace edge detector use only one kernel. To detect the edges of an image, this kernel detects 2nd order derivatives of image's intensity levels by using only single pass.
- "kernel 2" for detects edges with diagonals. It will give better approximation. Also, Laplace method gives faster calculations than others.

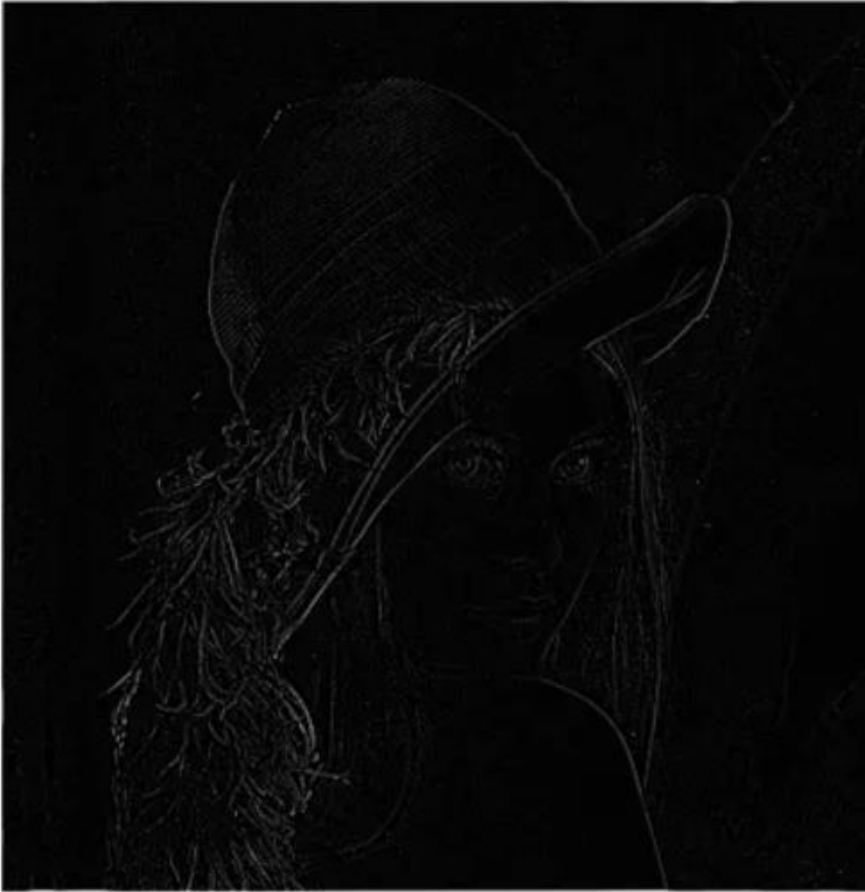
0	-1	0
-1	4	-1
0	-1	0

Kernel 1

-1	-1	-1
-1	8	-1
-1	-1	-1

Kernel 2

Laplacian Filter



Kernel 1



Kernel 2

Gaussian Filter

- This filter is a 2-D convolutional operator. It use to blur images.
- It removes details and noises.
- Gaussian filter is similar to mean filter. But main difference is, Gaussian filter use a kernel.
- That kernel has a shape of gaussian hump. Gaussian kernel weights pixels at its center much more strongly than its boundaries.
- There are different gaussian kernels. Based on the kernel size, output image will be different.

Gaussian Filter

$1/16$

1	2	1
2	4	2
1	2	1

3x3 kernel

$1/173$

1	4	7	4	1
4	16	26	16	4
7	26	41	26	7
4	16	26	16	4
1	4	7	4	1

5x5 kernel

Gaussian Filter



3X3 Kernel



5x5 Kernel

BOX Filter

- Box filter is a spatial domain linear image filter. Also, this box filter is a low-pass filter.
- It's operations are similar to average filtering technique



Average Filter



Box Filter

Image Segmentation

- Image segmentation involves partitioning a digital image into multiple regions (sets of pixels) to simplify the image or make it more meaningful and easier to analyze
- Segmentation is typically used to locate objects and boundaries (lines, curves, etc.) in images.

Image Segmentation

1. Medical Imaging:

Tumour detection, organ segmentation, and other diagnostic purposes.

2. Autonomous Vehicles:

Object detection, road segmentation, and obstacle avoidance.

3. Satellite and Aerial Imaging:

Land cover classification, urban planning, and environmental monitoring.

4. Face Recognition:

Facial feature extraction, emotion detection, and identity verification.

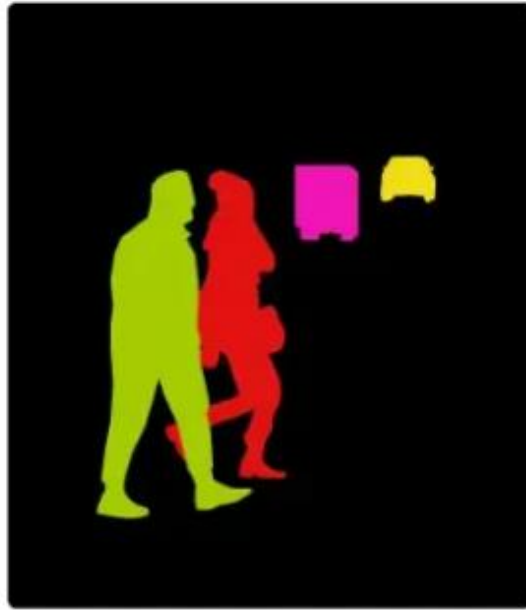
5. Agriculture:

Crop monitoring, disease detection, and yield estimation.

Image Segmentation - Types



**SEMANTIC IMAGE
SEGMENTATION**



**INSTANCE
SEGMENTATION**



**PANOPTIC
SEGMENTATION**

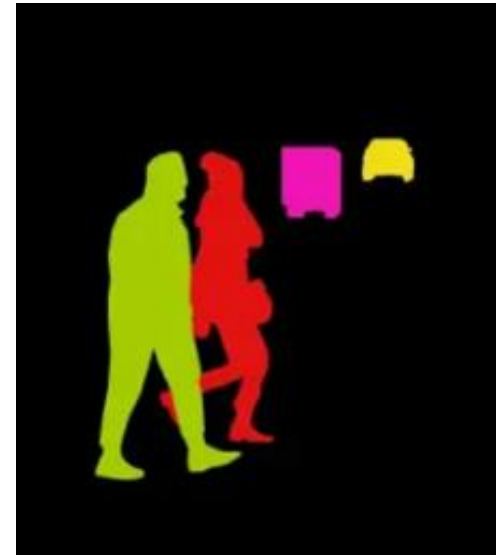
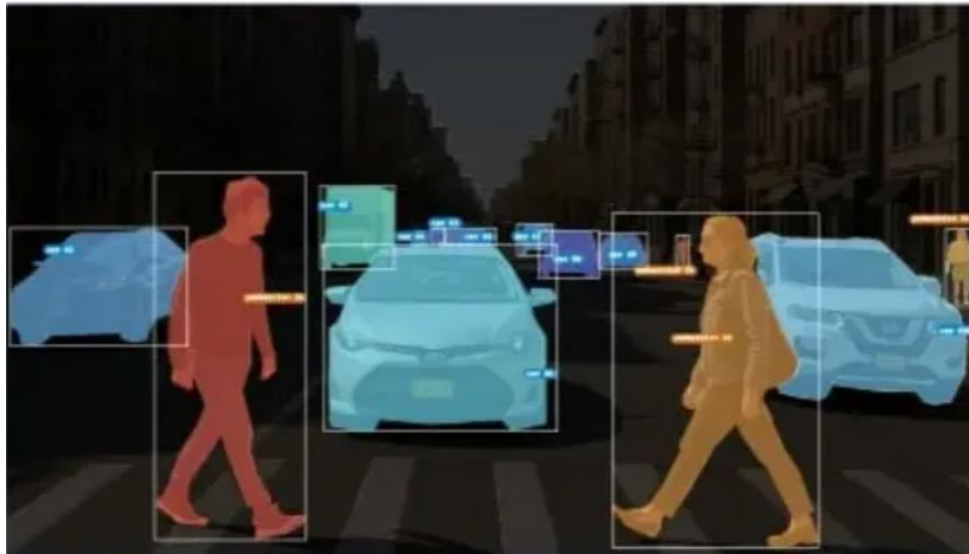
Semantic Segmentation

- Semantic Segmentation involves assigning a class label to every pixel in an image based on shared characteristics such as colour, texture and shape.
- This method treats all pixels belonging to the same class as identical without distinguishing between individual objects.
- **For example:** In an image with multiple trees all pixels corresponding to any tree would be labelled as "tree" regardless of how many trees appear in the image



Instance Segmentation

- Instance Segmentation extends semantic segmentation by not only labelling colour of each pixel but also distinguishing between individual objects of the same class.
- This approach identifies each object of the same class as a unique instance.
- **For example:** If there are multiple cats in an image it will differentiate each cat and treat them as separate entities despite them all being labelled as "cats."



Panoptic Segmentation

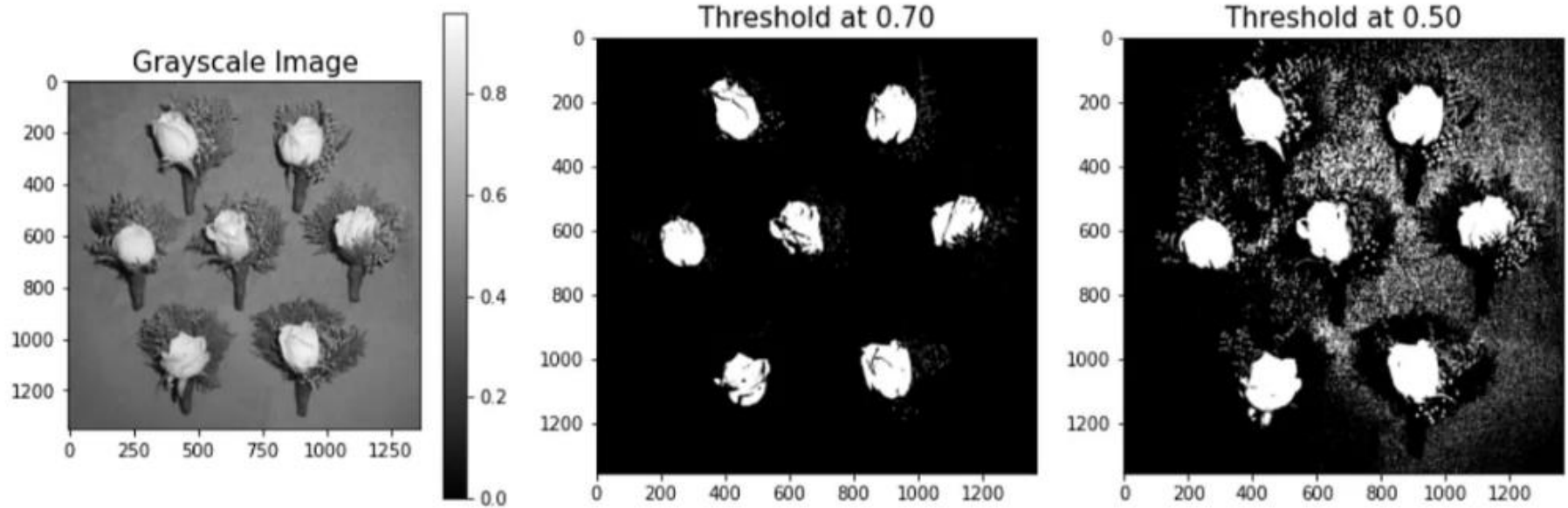
- Panoptic segmentation combines both semantic and instance segmentation techniques providing a complete image analysis.
- It assigns a class label to every pixel and also detects individual objects. This combines both broad categories and detailed object boundaries simultaneously.
- For example: In a traffic scene it would label all pedestrians and cars which is semantic segmentation while also outlining the location of each individual person and car which is instance segmentation



Image Segmentation Techniques

- **Thresholding:** Thresholding based image segmentation selects a threshold value and classifies image pixels between foreground and background based on intensity values.
- **Edge Detection:** Abrupt change in intensity or discontinuation in the image can be identified using Edge Detection. It uses algorithms like Sobel or Laplacian edge detectors.
- **Region Based Segmentation:** Region Based Segmentation method segments the image into smaller regions and iteratively merges them based on predefined attributes to handle noise in the image.
- **Clustering Algorithm:** Clustering algorithms like K-means or Gaussian models to group object pixels in an image into clusters based on similar features.
- **Watershed Segmentation:** Images are treated using Watershed Algorithm where the watershed lines are identified based on pixel intensity and connectivity.

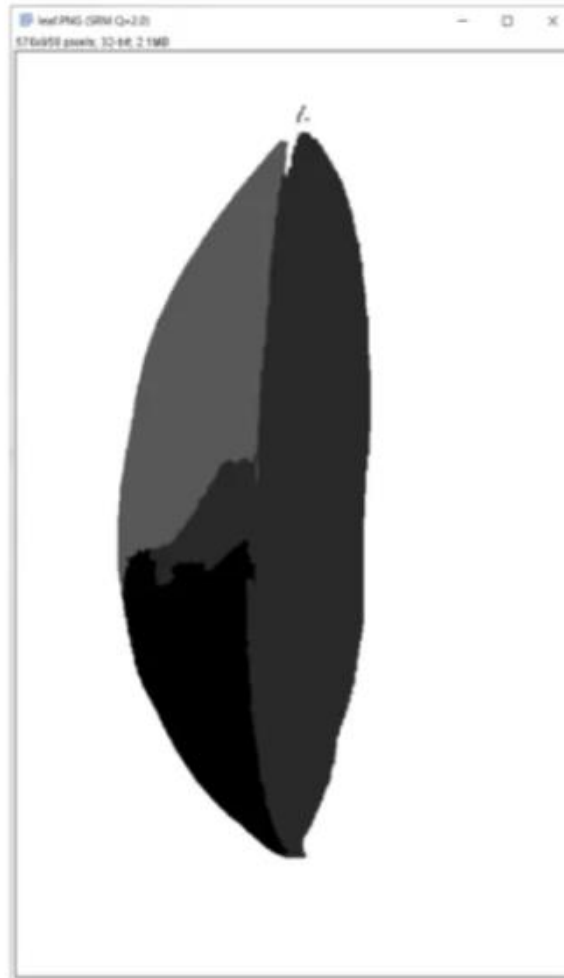
1. Thresholding



2. Edge Detection



3. Region Based Segmentation



4. Clustering

Original Image



Segmented Image when $K = 5$



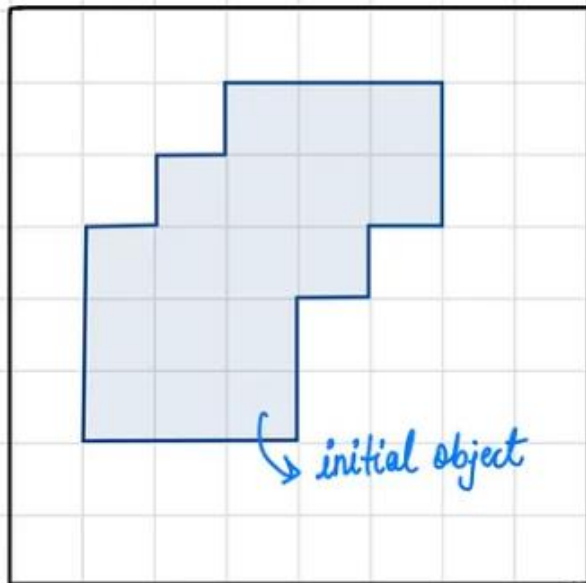
Morphological Operations

- **Morphological operations** are image processing techniques that focus on the **shape and structure of objects** in an image.
 - Erosion
 - Dilation
 - Opening
 - Closing

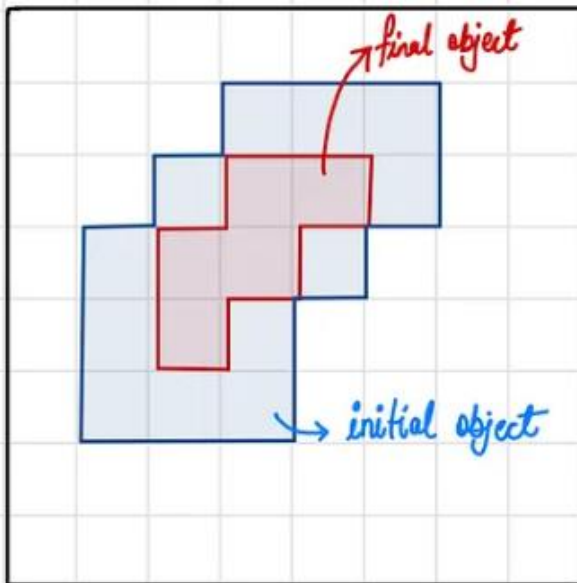
Erosion

- Erosion shrinks the image pixels, or erosion removes pixels on object boundaries.
- First, we traverse the structuring element over the image object to perform an erosion operation
- The output pixel values are calculated using the following equation.
Pixel (output) = 1 {if FIT}
Pixel (output) = 0 {otherwise}

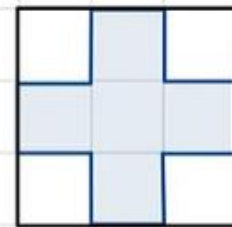
Erosion



Input image



Output image



Structuring Element



Erosion

Properties:

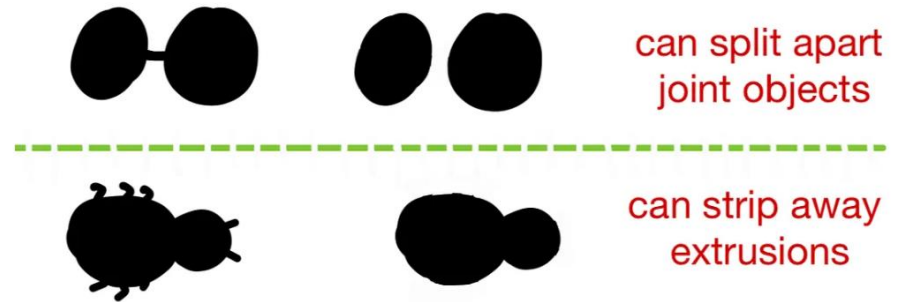
- It can split apart joint objects
- It can strip away extrusions

Effect:

- Removes small white regions
- Reduces object boundaries
- Eliminates small noise

Applications:

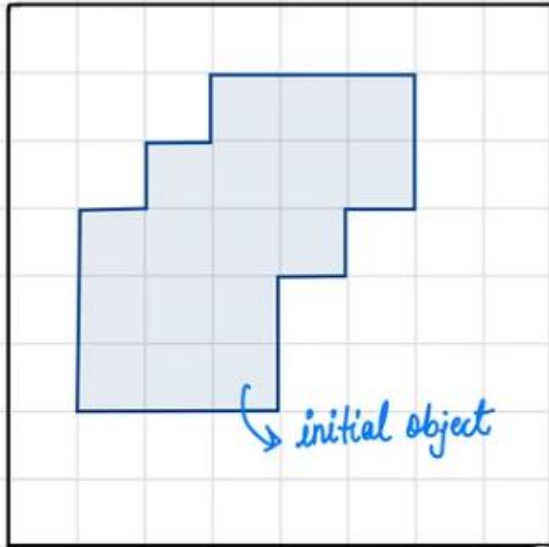
- Removing small objects
- Separating connected objects



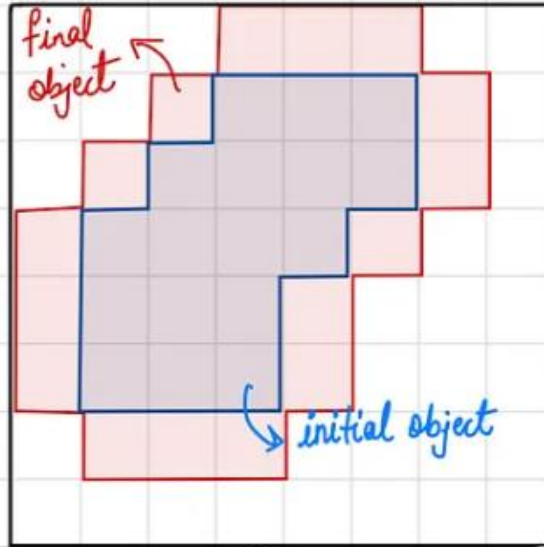
Dilation

- Dilation expands the image pixels, or it adds pixels on object boundaries.
- First, we traverse the structuring element over the image object to perform an dilation operation
- The output pixel values are calculated using the following equation.
Pixel (output) = 1 {if HIT}
Pixel (output) = 0 {otherwise}

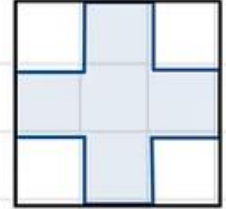
Dilation



Input image



Output image



Structuring Element

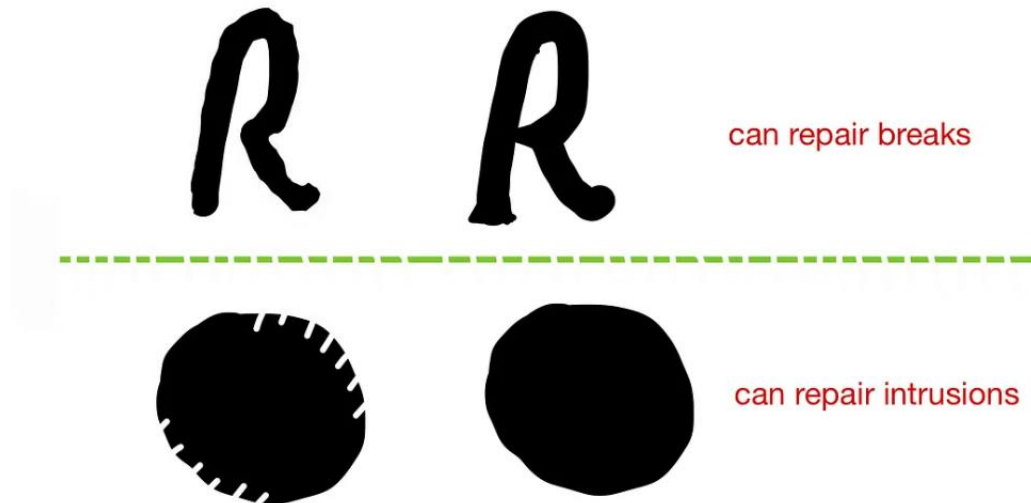
A

A

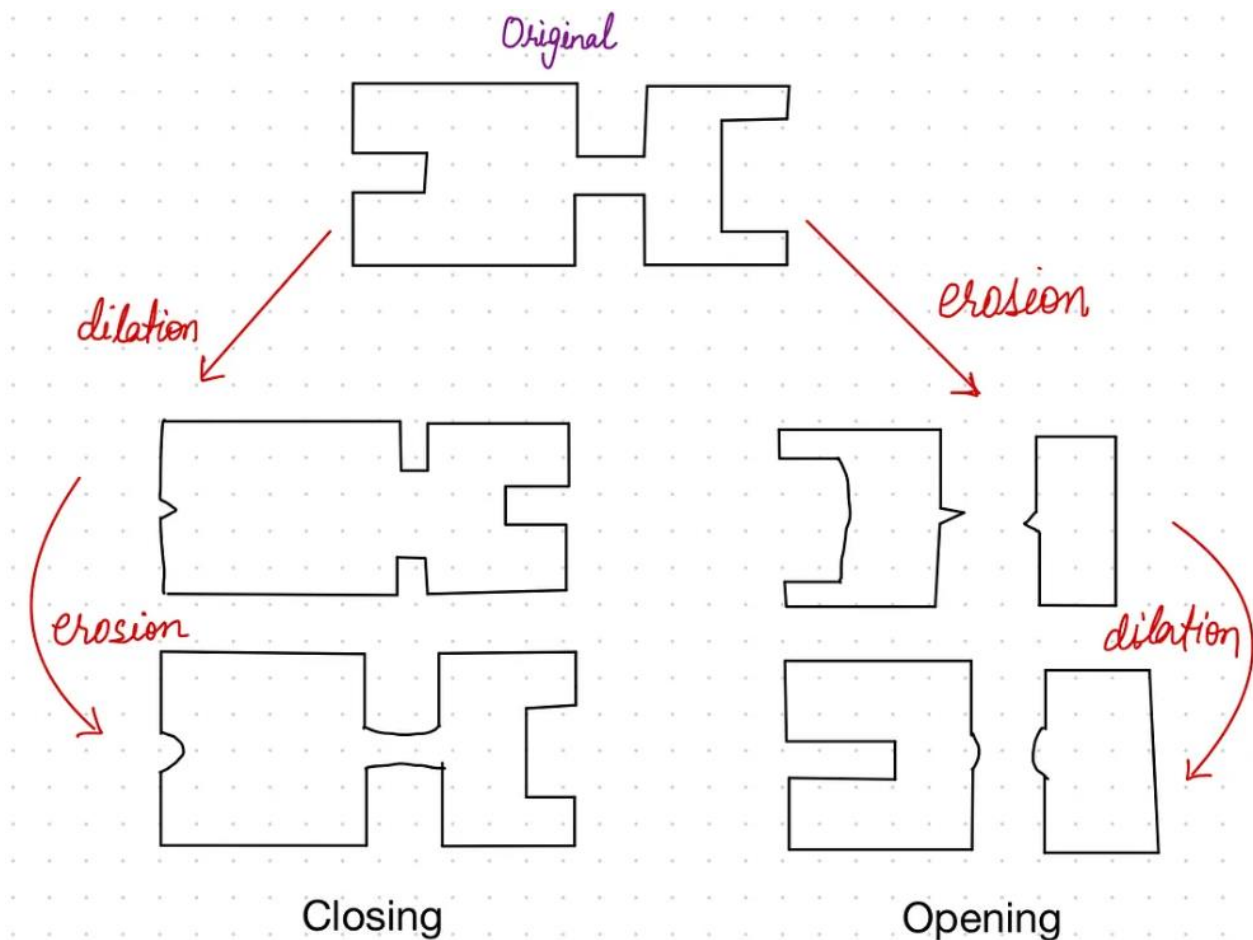
A

Dilation

- **Properties:**
 - It can repair breaks
 - It can repair intrusions
- **Effect:**
 - Fills small holes
 - Connects nearby objects
 - Expands boundaries



Opening and Closing



Opening

Opening = Erosion followed by Dilation

$$A \circ B = (A \ominus B) \oplus B$$

Effect:

- Removes small objects
- Smooths boundaries
- Breaks narrow connections

Closing

Closing = Dilation followed by Erosion

$$A \bullet B = (A \oplus B) \ominus B$$

Effect:

- Fills small holes
- Connects close objects
- Smooths contours

Morphological Operations

Original Image



Erosion



Dilation



Opening



Closing



Morphological Operations

Original image



Dilated image



Eroded image



Opened image



Closed image

